Investigation: FORJAS TAURUS SA (CVM code 6173), fiscal year 2012, as it was on 2013-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Metalurgia e Siderurgia (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

Document used

2012 DFP, version 3, filed on 2013-04-24; origin: ENET; type: consolidated; original scale: —.
Versions known on the date: 1 (2013-03-26); 2 (2013-04-04); 3 (2013-04-24).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=26284
- Structure read from a document filed after the date (study 1 behaviour, amendment 8)
- ENET version: the cache keeps values already in reais and did not record the page's scale (fetches before 2026-11-09)

Possible manipulation (engine v1, study 1)

Beneish M-score
  Value: −2.002 (index)
  Alert by the study 1 rule: no
  Rule: M > −1.78 (pre-registration, section 4)
  Percentile in sector and year: 91 (n = 16)
  Percentile in group and year: 81 (n = 334)
  The company's own history: fewer than 5 previous fiscal years with a value
  Confidence in the data: medium

DSRI, days' sales in receivables index
  Value: 0.931 (index)
  M-score component; it has no rule of its own
  Percentile in sector and year: 17 (n = 21)
  Percentile in group and year: 33 (n = 412)
  The company's own history: fewer than 5 previous fiscal years with a value
  Confidence in the data: medium

GMI, gross margin index
  Value: 1.096 (index)
  M-score component; it has no rule of its own
  Percentile in sector and year: 55 (n = 21)
  Percentile in group and year: 69 (n = 426)
  The company's own history: fewer than 5 previous fiscal years with a value
  Confidence in the data: medium

AQI, asset quality index
  Value: 2.039 (index)
  M-score component; it has no rule of its own
  Percentile in sector and year: 88 (n = 21)
  Percentile in group and year: 97 (n = 421)
  The company's own history: fewer than 5 previous fiscal years with a value
  Confidence in the data: medium

SGI, sales growth index
  Value: 1.134 (index)

M-score component; it has no rule of its own
Percentile in sector and year: 88 (n = 21)
Percentile in group and year: 55 (n = 427)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 1.005 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 53 (n = 16)
Percentile in group and year: 65 (n = 351)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.839 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 17 (n = 21)
Percentile in group and year: 19 (n = 420)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.015 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 69 (n = 21)
Percentile in group and year: 65 (n = 438)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 1.027 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 55 (n = 21)
Percentile in group and year: 53 (n = 432)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 2.7 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 27.2 days)
Percentile in sector and year: 68 (n = 20)
Percentile in group and year: 62 (n = 396)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 15.44 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 69 (n = 21)

Percentile in group and year: 75 (n = 394)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.021 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 69 (n = 21)
Percentile in group and year: 64 (n = 437)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 2.919 (index)
Components: X1 = 0.233; X2 = 0.110; X3 = 0.091; X4 = 0.400 (indices)
Percentile in sector and year: 75 (n = 22)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 5/9 (8 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 1
- Change in ROA: 0
- Cash above earnings (ACCRUAL): 1
- Change in leverage: 0
- Change in current ratio: 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin: 0
- Change in asset turnover: 1
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

What was known on the date

- DFP 2010, v1: 2011-03-30
- DFP 2010, v2: 2011-03-30
- DFP 2010, v3: 2011-04-12
- FCA 2012, v1: 2012-02-29
- DFP 2011, v1: 2012-03-27
- FCA 2012, v2: 2012-05-18

- FCA 2012, v3: 2012-06-11
- DFP 2012, v1: 2013-03-26
- DFP 2012, v2: 2013-04-04
- DFP 2012, v3: 2013-04-24

DFP and FCA filed with the CVM by the reference date; ITR comes in a later phase.
Material facts and market notices filed by the date (IPE open data, since 2010): 7 material fact documents and 25 market notice documents.
Material facts filed in the 24 months before the date (from 2011-05-01 to 2013-04-30):
- 2013-01-07: Celebração ou Extinção de Contrato
- 2012-06-21: Alienação das Operações da Controlada TMFL
- 2012-05-02: subject omitted (possible person's name); see the document at the CVM
- 2011-10-13: Aquisição da totalidade do capital social de empresa metalúrgica
- 2011-07-08: Crédito de Ações aos Acionistas em Decorrência da Reestruturação
- 2011-05-10: Reestruturação Societária

Older material facts, the market notices and the link to each document are in the JSON and on screen.
Material facts and market notices filed with the CVM by the date, from the IPE open data (since 2010): filing date, category, subject (as filed, in Portuguese) and link. The content of the documents is not downloaded, read or summarized, and the list generates no alert, score or ranking. The counts are of documents: versions, resubmissions and English versions count separately; in the list, versions of the same document and documents with the same date, category and subject appear on a single line, with a link to each one.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

the open-data opinions section is only for the latest DFP version; the version used came from ENET
Auditor in the 2010 FCA: KPMG Auditores Independentes.
Auditor in the 2011 FCA: KPMG Auditores Independentes.
Auditor in the 2012 FCA: Ernst & Young Terco Auditores Independentes S/S, KPMG Auditores Independentes.
Auditor changes (one line per change, canonical name by the firm's CVM code):
- 2012: entry of Ernst & Young Terco Auditores Independentes S/S, with no other firm leaving.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

Document excerpts (copied; no alert)

Excerpts copied from the documents, with the page, and the Reference Form table. It is not a signal family: it generates no alert, score or ranking. Absence of mention does not prove absence.
How the reader was tested: 22 held-out cases, a single measurement with the frozen reader (dd688da): precision 0.973, coverage 0.915, literal check on 146 of 146 excerpts, no natural-person name (docs/leitor_medida_f1_2.json).

Explanatory notes · Related parties
Note: 22. Partes relacionadas (pages 63 to 65)
- [parent company; subsidiaries] “Efeito de resultado das transações das controladas com a Saldos das controladas em aberto com a controladora controladora Ativo não circulante Ativo (créditos com Total Passivo circulante pessoas de circulante Total de (Clientes) ligadas)

ativo (*) passivo Receita Despesa 31/12/2011” (p. 63, original excerpt in Portuguese)
- [parent company; subsidiaries] “(*) Registrados como contas a pagar (**) Alienação de imobilizado da controladora para a controlada.” (p. 63, original excerpt in Portuguese)
- [subsidiaries] “As operações envolvendo a Companhia e a controlada Taurus Holdings, Inc., referem-se a vendas de armas para serem comercializadas pela controlada no mercado americano.” (p. 63, original excerpt in Portuguese)
- [subsidiaries] “Em relação às operações envolvendo a Companhia e a controlada Polimetal Metalurgia e Plásticos Ltda., referem-se a compra de produtos em processo, uma vez que a controlada realiza uma parte do processo produtivo do segmento de armas.” (p. 63, original excerpt in Portuguese)
- [parent company; subsidiaries] “A controlada Taurus Máquinas-Ferramenta Ltda. possui contrato de mútuo a pagar com a controladora” (p. 64, original excerpt in Portuguese)
- Excerpt omitted; see page 64 of the document
- [parent company] “Ltda. possui contrato de mútuo com a controladora Forjas Taurus S.A. no valor de R$ 547 em 2012 (R$” (p. 64, original excerpt in Portuguese)
- [key management personnel] “Remuneração de pessoal-chave da administração” (p. 64, original excerpt in Portuguese)
- [key management personnel] “Em 2012, a remuneração do pessoal-chave da administração totalizou R$ 19.810 (R$ 26.894 em 2011), e” (p. 64, original excerpt in Portuguese)
- [key management personnel] “Remuneração de pessoal-chave da administração compreende:” (p. 64, original excerpt in Portuguese)
- [key management personnel] “Remuneração e benefícios dos diretores” (p. 64, original excerpt in Portuguese)
- [key management personnel] “estatutários e Conselho de Administração 10.268 14.968 10.268 14.968” (p. 64, original excerpt in Portuguese)
- [key management personnel] “Remuneração de pessoal-chave 9.542 11.926 4.643 7.150” (p. 64, original excerpt in Portuguese)
- [key management personnel] “Total 19.810 26.894 14.911 22.118” (p. 64, original excerpt in Portuguese)
- [key management personnel] “A Companhia não possui políticas de benefícios de remuneração ao pessoal-chave da administração que se caracterizem como: benefícios pós-emprego, benefícios de rescisão de contrato de trabalho, remuneração com base em ações ou outros benefícios de longo prazo.” (p. 64, original excerpt in Portuguese)
- [key management personnel] “Operações de diretores e pessoal-chave da administração” (p. 65, original excerpt in Portuguese)
- [key management personnel] “Os diretores e pessoal-chave da administração controlam diretamente 44,18% das ações com direito de voto da Companhia.” (p. 65, original excerpt in Portuguese)
- [subsidiaries] “Os empréstimos e financiamentos estão garantidos por notas promissórias, alienação fiduciária de máquinas e equipamentos e hipoteca de imóveis. Os avais concedidos pela Companhia à suas controladas estão demonstrados a seguir:” (p. 65, original excerpt in Portuguese)

Fields of item 18 of CPC 05 (R1)
- [amount of the transactions; balances] “Efeito de resultado das transações das controladas com a Saldos das controladas em aberto com a controladora controladora Ativo não circulante Ativo (créditos com Total Passivo circulante pessoas de circulante Total de (Clientes) ligadas) ativo (*) passivo Receita Despesa 31/12/2011” (p. 63, original excerpt in Portuguese)
- [balances] “(*) Registrados como contas a pagar (**) Alienação de imobilizado da controladora para a controlada.” (p. 63, original excerpt in Portuguese)
- [amount of the transactions] “As operações envolvendo a Companhia e a controlada Taurus Holdings, Inc., referem-se a vendas de armas para serem comercializadas pela controlada no mercado americano.” (p. 63, original excerpt in Portuguese)
- [amount of the transactions] “Em relação às operações envolvendo a Companhia e a controlada Polimetal Metalurgia e Plásticos Ltda., referem-se a compra de produtos em processo, uma vez que a controlada realiza uma parte do processo produtivo do segmento de armas.” (p. 63, original excerpt in Portuguese)

- [terms and conditions] “As transações realizadas com partes relacionadas observam condições de preços e prazos efetuados de acordo com o pactuado entre as partes, e dentro das condições usuais de mercado, segundo a avaliação da Administração.” (p. 64, original excerpt in Portuguese)
- [balances] “A controlada Taurus Máquinas-Ferramenta Ltda. possui contrato de mútuo a pagar com a controladora” (p. 64, original excerpt in Portuguese)
- [terms and conditions] “527 em 2011). Os contratos não tem atualização monetárias ou taxas de juros.” (p. 64, original excerpt in Portuguese)
- [amount of the transactions] “Em 2012, a remuneração do pessoal-chave da administração totalizou R$ 19.810 (R$ 26.894 em 2011), e” (p. 64, original excerpt in Portuguese)
- [guarantees] “Avais entre partes relacionadas” (p. 65, original excerpt in Portuguese)
- [guarantees] “Os empréstimos e financiamentos estão garantidos por notas promissórias, alienação fiduciária de máquinas e equipamentos e hipoteca de imóveis. Os avais concedidos pela Companhia à suas controladas estão demonstrados a seguir:” (p. 65, original excerpt in Portuguese)

Market terms
- “As transações realizadas com partes relacionadas observam condições de preços e prazos efetuados de acordo com o pactuado entre as partes, e dentro das condições usuais de mercado, segundo a avaliação da Administração.” (p. 64, original excerpt in Portuguese)

The sentence about market terms is recorded as a fact, without proof (CPC 05 (R1), item 23).

Explanatory notes · Reverse factoring and forfaiting
The reader found no mention of reverse factoring (risco sacado), forfaiting or supplier agreements in the notes of this DFP. Absence of mention does not prove absence of the operation.

Auditor's report
Type of opinion: unqualified
- [emphasis of matter] “Ênfase Conforme descrito na nota explicativa 3.a, as demonstrações financeiras individuais foram elaboradas de acordo com as práticas contábeis adotadas no Brasil. No caso da Forjas Taurus S.A. essas práticas diferem do IFRS, aplicável às demonstrações financeiras separadas, somente no que se refere à avaliação dos investimentos em controladas e coligadas pelo método de equivalência patrimonial, enquanto que para fins de IFRS seria custo ou valor justo. Nossa opinião não está ressalvada em função deste assunto.” (original excerpt in Portuguese)
- [other matters] “Outros assuntos Demonstrações do valor adicionado Examinamos, também, as demonstrações individual e consolidada do valor adicionado (DVA), referentes ao exercício findo em 31 de dezembro de 2012, preparadas sob a responsabilidade da administração da Companhia, cuja apresentação é requerida pela legislação societária brasileira para companhias abertas e como informação suplementar pelas IFRS que não requerem a apresentação da DVA. Essas demonstrações foram submetidas aos mesmos procedimentos de auditoria descritos anteriormente e, em nossa opinião, estão adequadamente apresentadas, em todos os seus aspectos relevantes, em relação às demonstrações financeiras tomadas em conjunto.” (original excerpt in Portuguese)

Reference Form
The Reference Form for this fiscal year was not available on the date. It is the only version available in the open data; earlier versions are not there.

Limits of this section:
- In real documents, about 22% of the excerpts come out omitted by the name detector, including names of companies outside the registry.
- In tables, a header line (years, column titles) may still come out as an excerpt: in the reader's measurement, 1 of 19 header texts of the held-out cases came out as excerpts.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-

years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Structure read from a document filed after the date (study 1 behaviour, amendment 8)
- ENET version: the cache keeps values already in reais and did not record the page's scale (fetches before 2026-11-09)
- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a F1); leitor-f1-2 (commit dd688da). The generation date is in the demo manifest (demo/manifesto.json).

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: TAURUS ARMAS S.A.